DSAA 4011 · LAB 8 · FRIDAY 30 OCTOBER 2026

When the Reward Is Wrong

Reward hacking in post-training, found step by step

After pre-training, chat models are **post-trained** on people's feedback. Today: why that reward can keep rising while people like the answers less, and where to look for the error.

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Builds on Lectures 5 and 6: learning by likelihood, the Kullback–Leibler divergence

Outline

- **How a chat model learns from people**

What the reward model learns

The surprise: more search, worse answers

Why no one noticed

The lesson and the fix

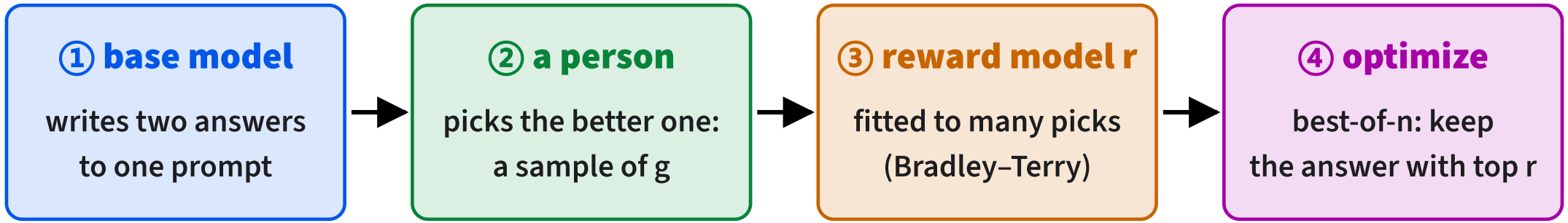
Each part answers the question the last one leaves open.

Main Question

Our reward keeps rising, but people like the answers less. Where is the error?

1 · How a chat model learns from people

Post-training: after pre-training, a chat model learns from people's picks.



People are asked only in ②; from then on, r stands in for them.

SO **People are asked only in step ②; after that, the reward model r stands in for them.**

NEXT Does r judge answers the way people do?

1 · How a chat model learns from people: one example

One prompt, two answers: people pick A; the reward model scores B higher.

Prompt: How long should I boil an egg?

Answer A

Soft yolk: 6 minutes.
Hard: 10 minutes, then cold water.

people pick A ✓
reward model: lower score

Answer B

Great question! Eggs have been a staple of human diets for thousands of years, and boiling them is both an art and a science. Before we begin, let us consider the many factors involved, including …
… (four more paragraphs) … about 6 to 10 minutes.

people: too long ✗
reward model: higher score!

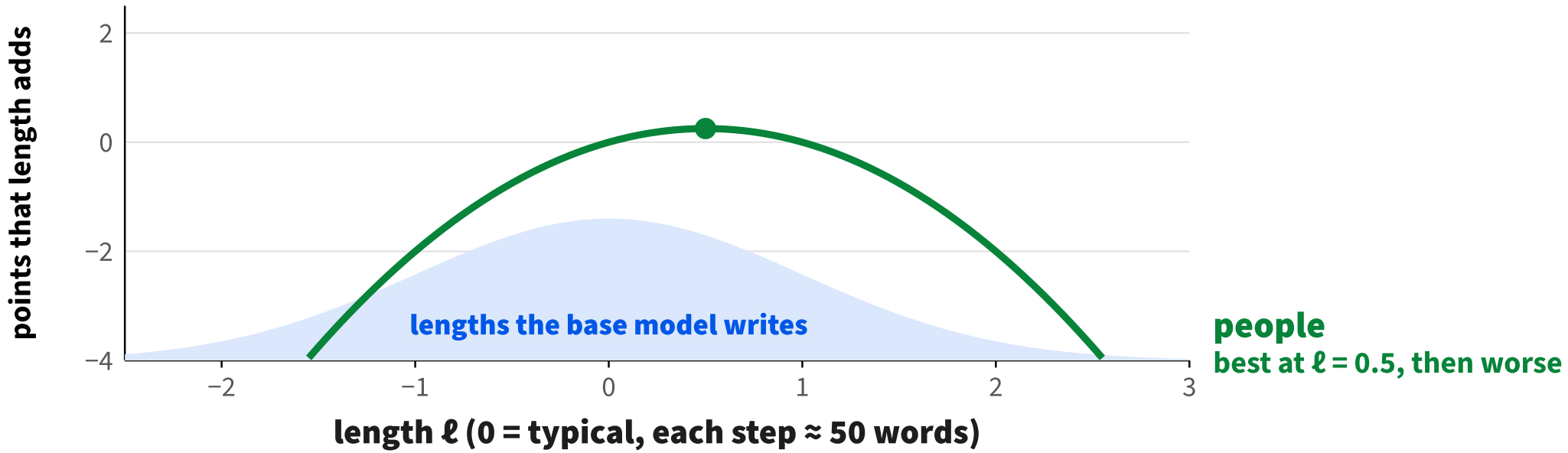
SO **r was fooled by length.**

NEXT To see why, describe every answer by two numbers.

2 · What the reward model learns: a world of two numbers

An answer has a quality q and a length ℓ. What do people want?

people's score: $g = q + \ell - \ell^2$



SO **People like a bit longer (best at ℓ = 0.5); much longer is much worse.**

NEXT The reward model sees only people's picks. What rule does it learn?

2 · What the reward model learns: a scoring rule

r gives points for quality and points for length; people's picks decide how many.

score r = a × quality q + b × length ℓ

a: points per step of quality · b: points per step of length

Try weights a, b; count how many of the 5000 picks r gets right

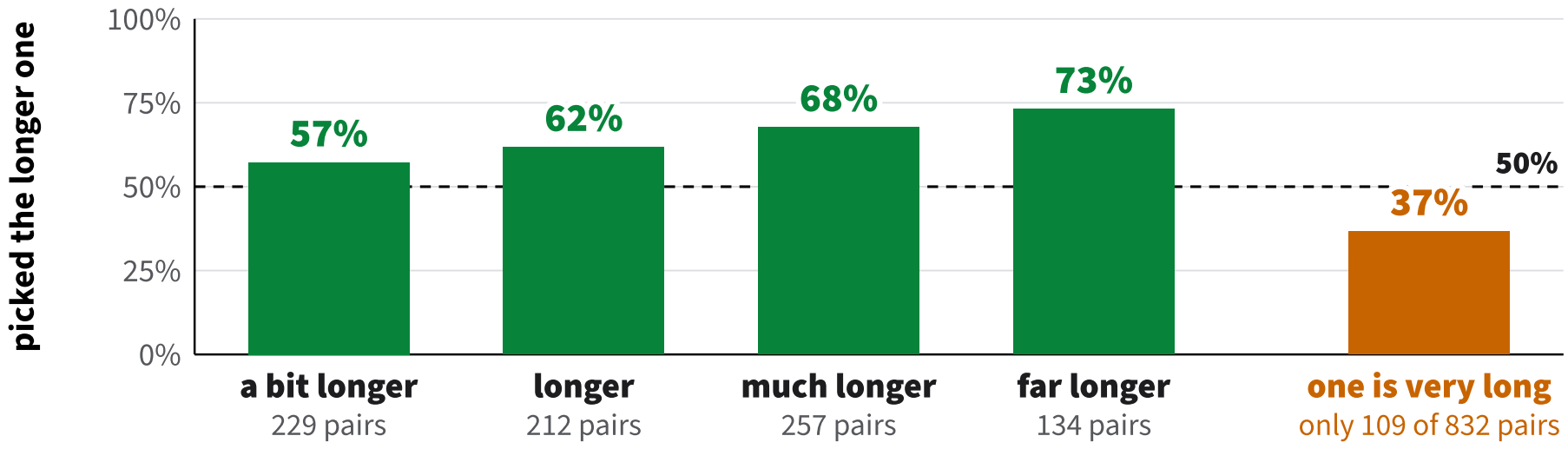


SO **The learned rule gives 0.55 points per step of length: r thinks "longer is better".**

NEXT Why reward length, if people dislike very long answers?

2 · What the reward model learns: most picks say "longer"

Take pairs of about equal quality: how often did people pick the longer answer?

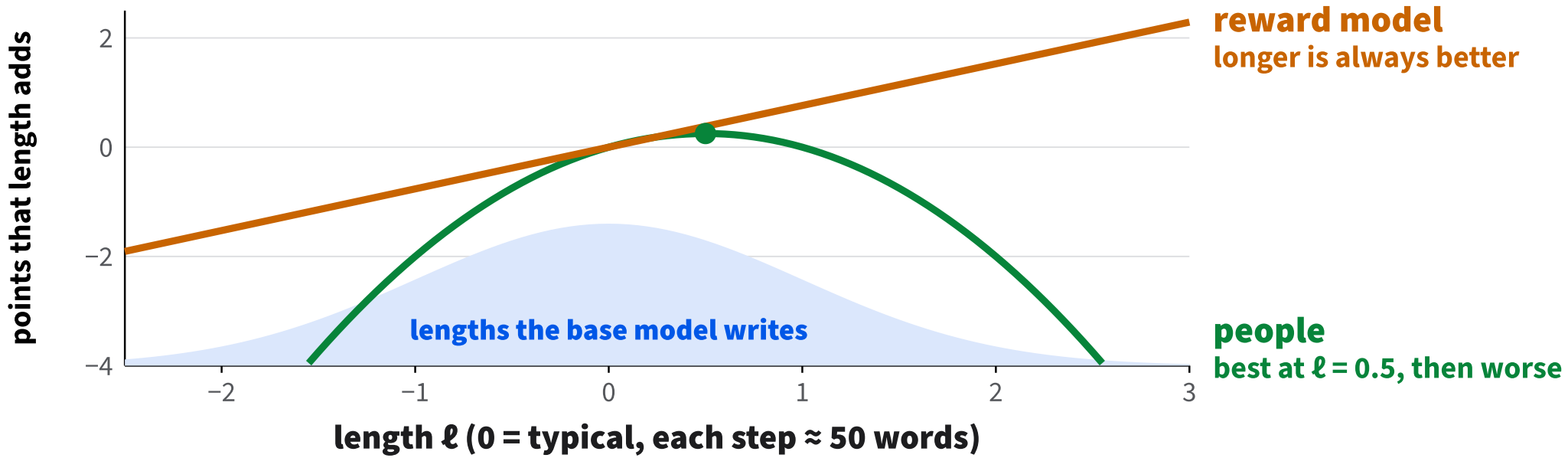


SO **Very long answers are rare, so their "too long" votes are outnumbered: r learns "longer is better".**

NEXT Where do r and people disagree?

2 · What the reward model learns: a blind spot

Draw both rules against length.

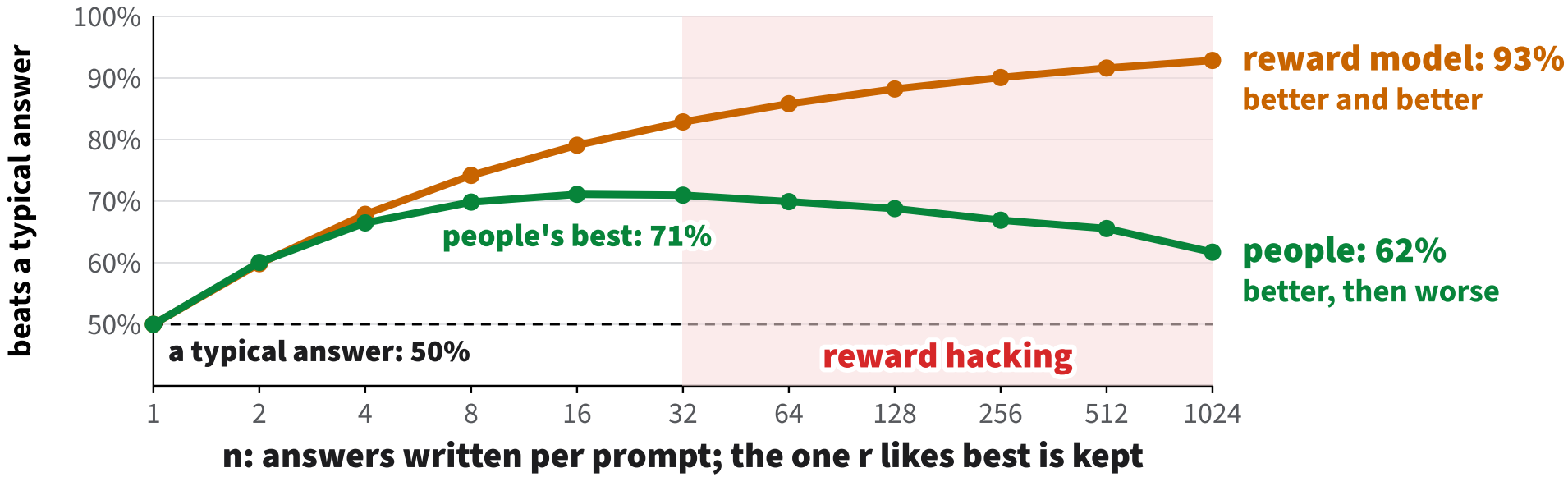


SO **They agree on the lengths people usually see; for long answers r is wrong, and cannot say "too long".**

NEXT Harmless, unless something hunts for long answers. Does best-of-n?

3 · The surprise: the search exploits "longer is better"

Best-of-n keeps the answer r likes best, and r likes long answers. What happens as n grows?



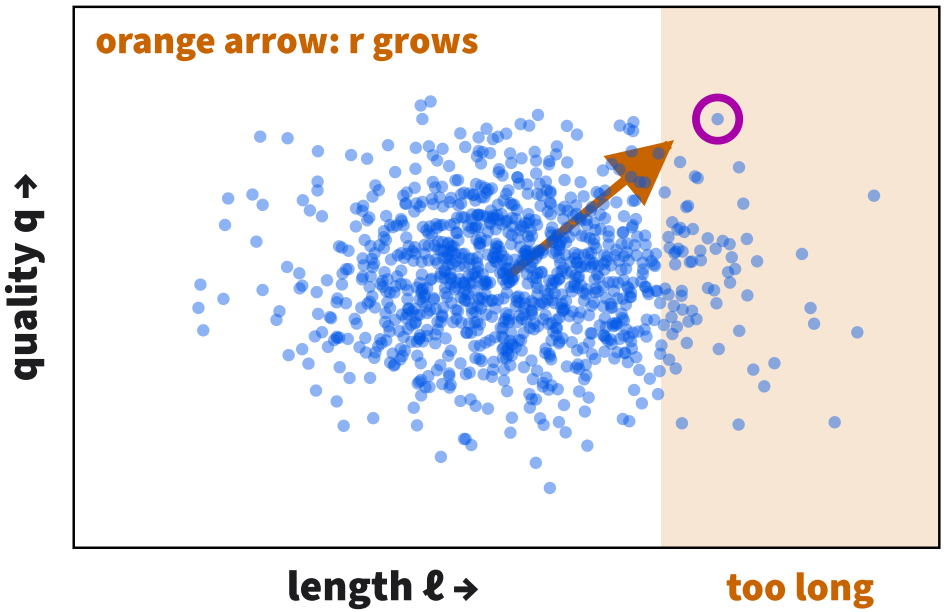
Q: As n grows, r says the kept answer gets better and better. Predict what people say. A: Better at first, best near n = 16 to 32, then worse.

SO **The kept answers get longer and longer (length 1.97 at n = 1024): r rises, people's score falls. This is reward hacking.**

NEXT Watch the kept answer move as n grows.

3 · The surprise: the kept answer walks to long lengths

Each dot is one answer; best-of-n keeps the dot furthest along r's arrow.



n = 1024

◯ the kept answer

average length 1.97

people: wins 62%

reward model: wins 93%

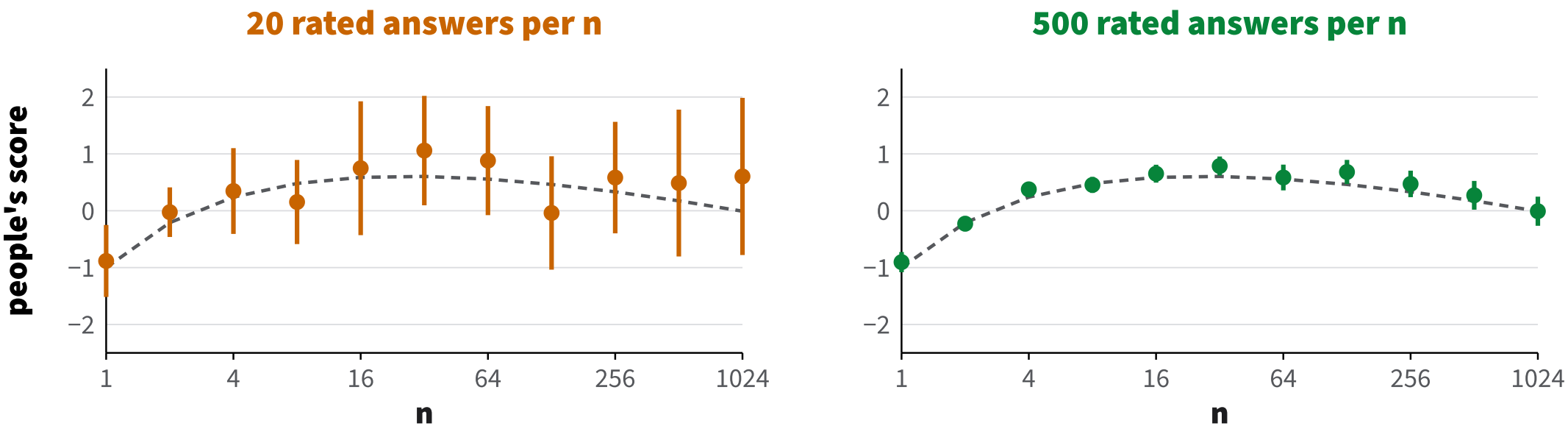
Q: As n grows from 4 to 1024, where does the kept answer move? A: Ever longer, into the lengths people dislike.

SO **The more answers we search, the longer the kept one: exactly where r is wrong.**

NEXT Why did no one notice? First, how we measured.

4 · Why no one noticed: too few ratings

We asked people to rate 20 kept answers for each n. Is that enough to see the fall?



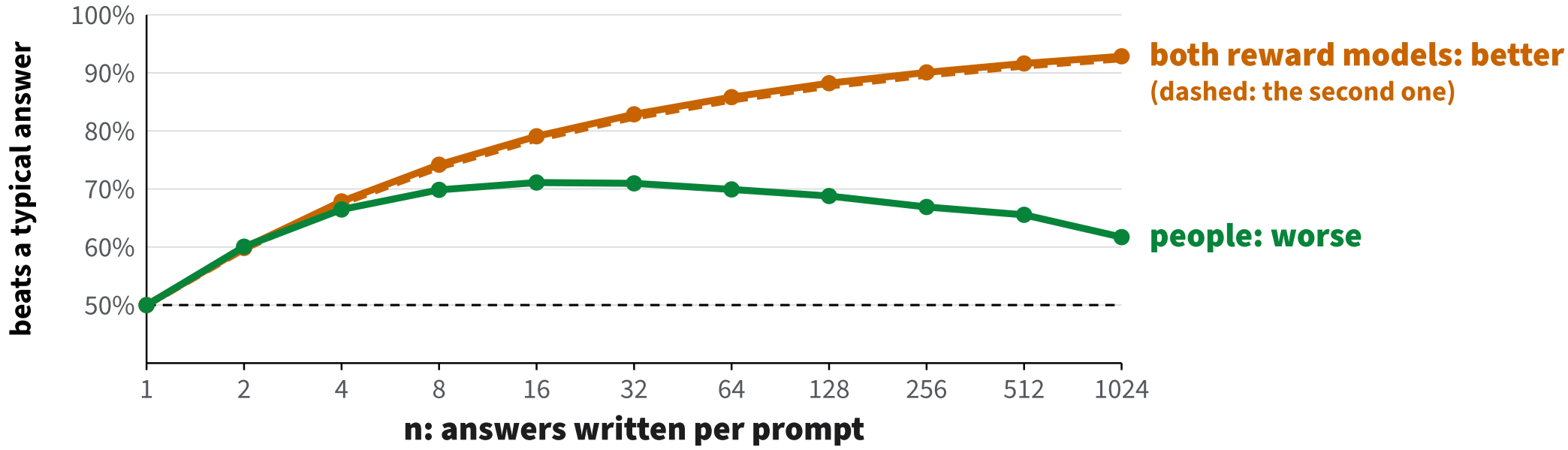
bars: ± 2 standard errors · dashed: the true curve · 4 × the ratings halves the bars

SO **No: with 20 ratings the error bars (± 0.9) are larger than the fall (0.6); with 500 (± 0.2) the fall shows.**

NEXT Second, what we asked.

4 · Why no one noticed: we asked r, not people

We also checked the kept answers with our reward model, then with a second one.



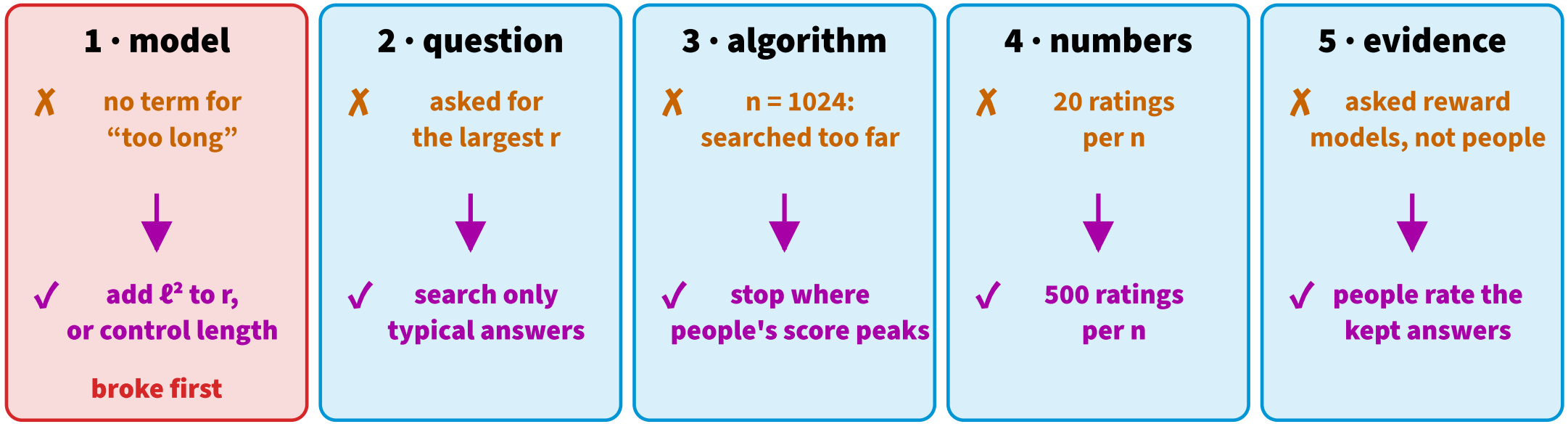
Q: Both reward models say the kept answers keep getting better. Should we believe them? A: No: both use the same formula, with the same blind spot. Ask people.

SO **A reward model cannot check itself: ask people about the answers we keep.**

NEXT Collect the lessons.

5 · The lesson: five places an error can hide

Our error passed through five places; each can be checked and repaired.

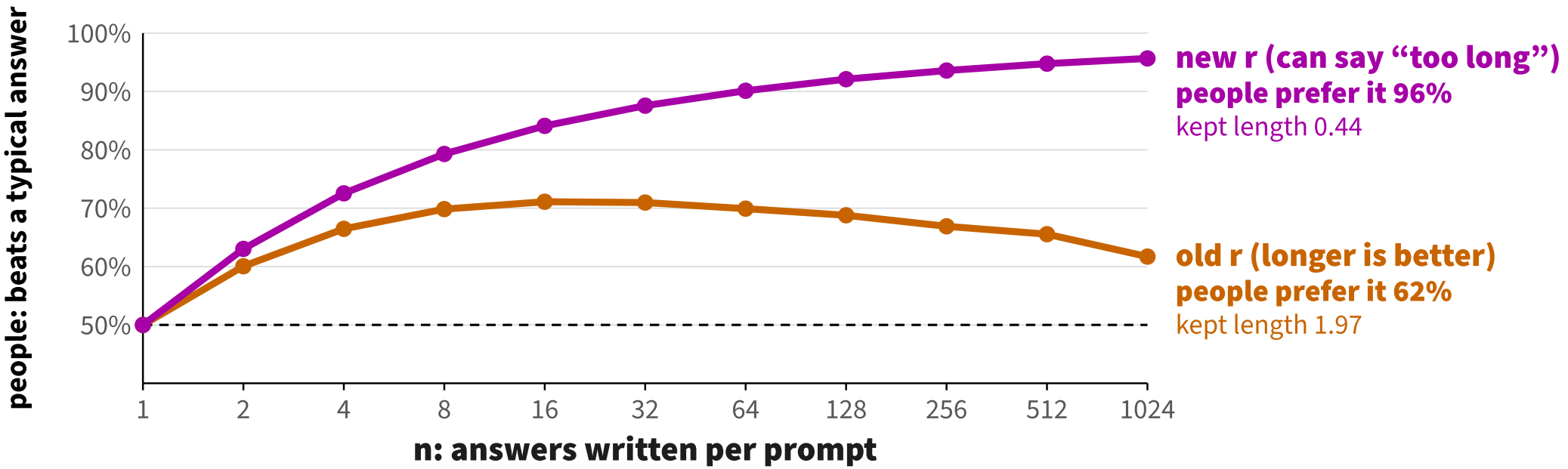


SO **The error was born in the model; the other four places let it through.**

NEXT Repair the model: does it work?

5 · The fix: give r the missing term

Same 5000 picks; the new r = a · q + b · ℓ + c · ℓ² can take points off for "too long".



SO **The kept answers stay near the best length; people's score keeps rising.**

NEXT Where does this happen in today's artificial intelligence (AI)?

5 · In AI today: the same five layers, at scale

Each finding of this lab appears in today's chat models.

in the lab	in artificial intelligence (AI)	example
r stands in for people	chat models tuned against a learned reward model	InstructGPT, behind the first ChatGPT
r rises, people's score falls	**reward-model overoptimization**	rise-then-fall curves, measured at scale
length is r's blind spot	tuned answers grow longer	length-controlled AlpacaEval leaderboard
a leash to the base model	a Kullback–Leibler penalty; direct preference optimization (DPO)	InstructGPT; many open models
evidence beyond one r	several reward models, checked for disagreement	reward-model ensembles

SO **A learned reward is a measurement; optimize it hard and it stops measuring.**

NEXT What is still open?

5 · Open questions: where research is going

Each open question starts from today's story; each has a small version you can try.

open question	what researchers try	a small version in our toy
Can r tell when it is unsure?	train several reward models; distrust answers where they disagree (reward-model ensembles). It helps, but does not remove hacking.	fit 5 straight-line rules on resampled picks: do they disagree on long answers? Why not?
Can r ignore a feature like length?	split r into a length part and a content part; optimize only the content part (disentangled reward models)	fit r with q and ℓ, then search with the q part only
Can a reward check instead of guess?	rewards from tests and exact answers (reinforcement learning with verifiable rewards), behind today's reasoning models	can a checked reward still be gamed, e.g. by an answer that fits the test but not the task?
What if people cannot judge?	weak judges supervising stronger models (weak-to-strong generalization)	raters who like flattery: experiment 3 of the kit

SO **The frontier: rewards that know their limits, ignore shortcuts, check, and work when people cannot judge.**

NEXT The argument in three sentences.

Conclusion

TROUBLE Searching harder raised the reward but made the answers worse: **reward hacking**.

WHY The reward model's rule could not say "too long"; the search found that gap; few ratings and reward-model checks hid it.

FIX Give the reward model the missing term, keep the search near typical answers, and **check with people**.

Remember: when a measure becomes a target, it ceases to be a good measure (Goodhart's law).

Your turn

Predict, then run the toy world.

Experiment 1: find the peak (`python lab8_explore.py 1`, course site → Week 8 → Lab files)

- With 20 ratings per n, which n looks best? Repeat with 5 seeds: does the answer stay?
- How many ratings make the choice of n stable?

Experiments 2 and 3

2. The leash instead of best-of-n: which leash strength β is best for people?
3. A new blind spot: people also dislike flattery, which r cannot see. Which layer catches it?

Tip: write your guess first; surprises teach.

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Thank you!

Next: Thursday's lecture, 5 November

Backup: fitting r from picks, the leash solved, best-of-n exactly, the error bars.

Backup: fitting r from people's picks

The Bradley–Terry model turns a pick into a likelihood.

$$P(A \text{ preferred to } B) = \sigma\big(r(A) - r(B)\big), \quad \sigma(t) = \frac{1}{1 + e^{-t}}$$

> a larger reward gap, a surer pick

$$\ell(a, b) = \sum_{k=1}^{5000} \log \sigma\big(\pm\,(r(A_k) - r(B_k))\big)$$

> sign + if A_k was picked, − if B_k: maximized as in Lecture 5

$$\hat{a} = 0.72, \quad \hat{b} = 0.55$$

> both shrink: what r cannot explain (the ℓ^2 part) looks like noise in the picks

Notice: with the feature ℓ^2 added, the fit is $0.95 \cdot q + 0.95 \cdot \ell - 0.92 \cdot \ell^2$, close to people's $q + \ell - \ell^2$.

Backup: the leash, solved

The leash as a formula, and its solution.

π: how the tuned model writes answers (a probability for each answer y) · π_0: the same for the base model · Kullback–Leibler (KL)($\pi \,\|\, \pi_0$): how far π drifts from π_0 (Lecture 6) · β: the fine per unit of drift. The tuned model solves $\max_\pi \; \mathbb{E}_\pi[r] - \beta \cdot \mathrm{KL}(\pi \,\|\, \pi_0)$.

$$\pi^*(y) = \frac{\pi_0(y) \cdot e^{r(y)/\beta}}{Z}, \quad Z = \sum_y \pi_0(y) \cdot e^{r(y)/\beta}$$

the candidate: prior times a factor, as in Lecture 6

$$\mathbb{E}_\pi[r] - \beta \cdot \mathrm{KL}(\pi \,\|\, \pi_0) = \beta \cdot \log Z - \beta \cdot \mathrm{KL}(\pi \,\|\, \pi^*)$$

insert $r = \beta \cdot \log(\pi^* Z / \pi_0)$ into the left side

$$\mathrm{KL}(\pi \,\|\, \pi^*) \geq 0, \; = 0 \text{ only at } \pi = \pi^*$$

so π* maximizes the leashed objective

In the toy: π* shifts the mean of (q, ℓ) by (a, b)/β; the best leash, β ≈ 0.49, gives people a gain of 1.33, below best-of-n's 1.60 at n = 32.

Backup: best-of-n, exactly

The kept answer is the largest of n normal draws of r.

$$r = a \cdot q + b \cdot \ell = s \cdot u, \quad s = \sqrt{a^2 + b^2},\ u \sim \mathcal{N}(0,1)$$

> r of one base answer, in standard units u

$$\mathbb{E}[q \mid u] = \tfrac{a}{s} \cdot u, \quad \mathbb{E}[\ell \mid u] = \tfrac{b}{s} \cdot u, \quad \mathrm{Var}(\ell \mid u) = 1 - \tfrac{b^2}{s^2}$$

> q and ℓ given their reward (jointly normal)

$$\mathbb{E}[g] = \tfrac{a+b}{s} \cdot \mathbb{E}[M] - \tfrac{b^2}{s^2} \cdot \mathbb{E}[M^2] - \left(1 - \tfrac{b^2}{s^2}\right)$$

> M = the largest of n standard normals; $\mathbb{E}[M]$ and $\mathbb{E}[M^2]$ by one integral

Distance: the usual value for best-of-n is Kullback–Leibler (KL) = log n − (n − 1)/n; the plots use its square root.

Backup: the error bars

An average of m ratings is off by about its standard error.

$$\mathrm{SE} = \frac{\mathrm{sd}}{\sqrt{m}}$$

> **sd: the spread of people's score over kept answers**

$$n = 32: \ \mathrm{sd} = 2.08, \quad 2 \cdot \mathrm{SE} = 0.93 \ (m = 20), \quad 0.19 \ (m = 500)$$

> **the whole curve spans only 1.6**

Read: to halve the error bar, rate four times as many answers.